$\boxed{T8}$ 2 партии по 100 деталей

$\eta \backslash \xi$	A_1	A_2	номера партии
"1"	25	52	$\frac{77}{200}$
"2"	50	41	$\frac{91}{200}$
"3"	25	7	$\frac{32}{200}$
	$\frac{100}{200}$	$\frac{100}{200}$	

разные

H_0: ξ и η независимы $\qquad \alpha = 0{,}05$

H_1: $\overline{H_0}$

$$\tilde{\Delta} = \frac{\left(25 - 200\cdot\frac{1}{2}\cdot\frac{77}{200}\right)^2}{200\cdot\frac{1}{2}\cdot\frac{77}{200}} + \frac{\left(52 - 200\cdot\frac{1}{2}\cdot\frac{77}{200}\right)^2}{200\cdot\frac{1}{2}\cdot\frac{77}{200}} +$$

$$+ \frac{\left(50 - 200\cdot\frac{91}{200}\cdot\frac{1}{2}\right)^2}{200\cdot\frac{1}{2}\cdot\frac{91}{200}} + \frac{\left(41 - 200\cdot\frac{91}{200}\cdot\frac{1}{2}\right)^2}{200\cdot\frac{1}{2}\cdot\frac{91}{200}} +$$

$$+ \frac{\left(25 - 200\cdot\frac{1}{2}\cdot\frac{32}{200}\right)^2}{200\cdot\frac{1}{2}\cdot\frac{32}{200}} + \frac{\left(7 - 200\cdot\frac{1}{2}\cdot\frac{32}{200}\right)^2}{200\cdot\frac{1}{2}\cdot\frac{32}{200}} =$$

$$\approx 20{,}48$$

$$\Delta \rightsquigarrow \chi^2((2-1)(3-1)) = \chi^2(2)$$

$$\text{p-value} = P(\Delta \geq \tilde{\Delta} \mid H_0) = \int_{20,48}^{+\infty} q(t)\,dt =$$

$$= \int_{20,48}^{+\infty} \frac{\left(\frac{1}{2}\right)^{\frac{k}{2}}}{\Gamma\left(\frac{k}{2}\right)} x^{\frac{k}{2}-1} e^{-\frac{x}{2}}\,dx = 3,57 \cdot 10^{-5} < \alpha = 0,05$$

$\Rightarrow$ Отвергаем H_0